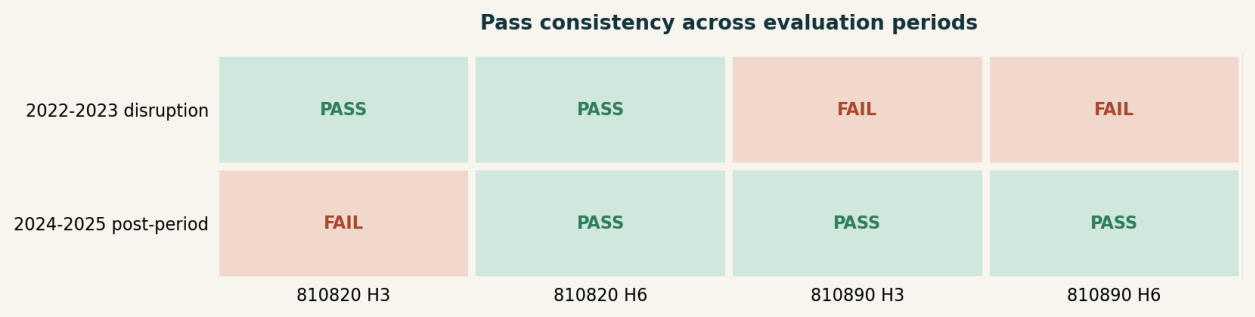
EVALUATION BRIEF

Titanium Model A: what the two tests prove

Required 2022-2023 disruption evidence, additional 2024-2025 post-disruption evidence, training design, limitations and production-readiness actions.

Bottom line: only HS 810820 H6 passed in both periods. The system is appropriate for shadow monitoring, but the complete H3/H6/H12 service is not yet production approved.

Evidence	Purpose	Result	What it tells us
2022-2023 disruption	Required stress test	2 of 4 pass	810820 adapted; 810890 did not beat the baseline reliably.
2024-2025 post-period	Additional generalisation check	3 of 4 pass	Recent performance improved, but 810820 H3 became inconsistent.
Combined	Consistency across regimes	1 of 4 passes both	810820 H6 is the only consistently supported forecast.



A PASS requires at least 90% P05-P95 coverage, MIS below the no-change baseline, complete outputs, and - in the disruption test - six-month rolling coverage recovery.

What these tests prove - in plain language

- The disruption test asks whether the model remains useful when the market changes abruptly.
- The post-period test asks whether performance continues after the main shock.
- Together they reveal stability across regimes; they do not prove future accuracy.
- The evidence supports 810820 H6 most strongly. Other outputs require analyst review.

REQUIRED TEST

2022-2023 disruption stress test

The specification freezes model choices before February 2022 and evaluates monthly forecast origins from February 2022 through December 2023. Models are retrained at each origin using only information available at that date. This simulates deployment during the Russian titanium disruption without retuning against the answers.

Series	Selected model	Coverage	MIS vs baseline	Recovery	Result
810820 H3	Elastic Net	100.0%	0.992 (0.8% better)	Yes	PASS
810820 H6	Random Forest	100.0%	0.989 (1.1% better)	Yes	PASS
810890 H3	Adaptive no-change	91.3%	1.048 (4.8% worse)	Yes	FAIL
810890 H6	Adaptive no-change	95.7%	1.051 (5.1% worse)	No	FAIL

Observations

- 810820 H3 and H6 achieved 100% coverage and recovered by July 2022.
- Their MIS advantages were small: 0.8% for H3 and 1.1% for H6.
- 810890 H3 covered enough outcomes but its MIS was 4.8% worse than baseline.
- 810890 H6 had adequate total coverage, but MIS was 5.1% worse and recovery failed.

Meaning: the model handled unwrought titanium and powders better than the heterogeneous processed-products category during the shock.

Why both coverage and MIS matter

Coverage checks whether the actual value falls inside the stated 90% interval. MIS also penalises intervals that are excessively wide or miss badly. A model cannot pass by issuing a range so wide that it becomes operationally useless.

ADDITIONAL EVIDENCE

2024-2025 post-disruption evaluation

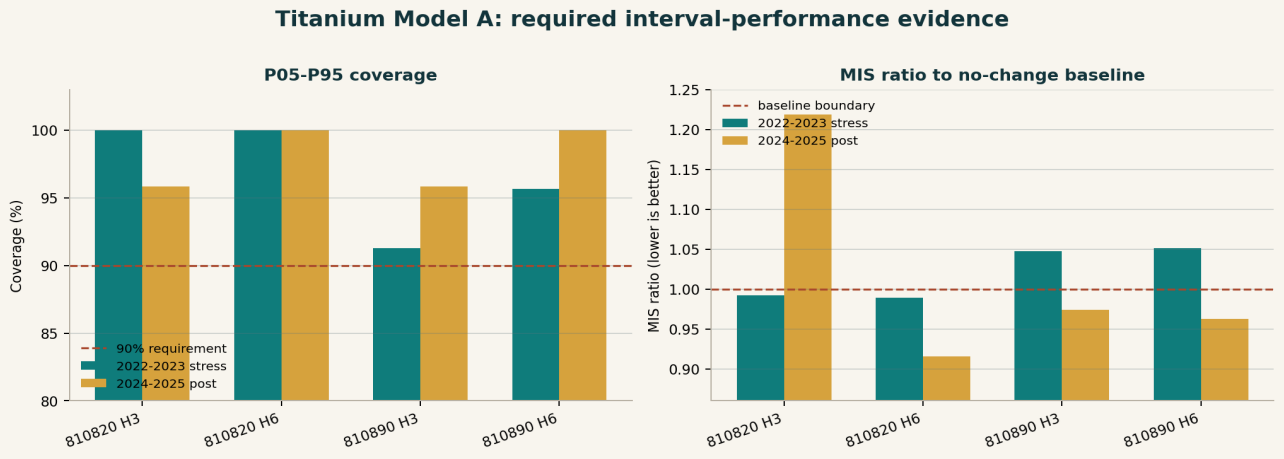
This period is not explicitly required by the specification. It was added to check whether the frozen model choices continued to work in a later regime. It is useful evidence, but not a pristine holdout because these data were explored during development.

Series	Selected model	Coverage	MIS vs baseline	Result
810820 H3	Elastic Net	95.8%	1.219 (21.9% worse)	FAIL
810820 H6	Random Forest	100.0%	0.916 (8.4% better)	PASS
810890 H3	Adaptive no-change	95.8%	0.974 (2.6% better)	PASS
810890 H6	Adaptive no-change	100.0%	0.963 (3.7% better)	PASS

Observations

- 810820 H3 failed because MIS was 21.9% worse than baseline despite 95.8% coverage.
- 810820 H6 remained strongest: 100% coverage and MIS 8.4% better than baseline.
- 810890 H3 and H6 passed, but gains over baseline were modest at 2.6% and 3.7%.
- The adaptive no-change method remained selected for 810890 H3/H6.

Meaning: recent results are better for 810890, but its failure under disruption means robustness has not been demonstrated. 810820 H3 also changes from pass to fail across regimes.



Dashed lines mark the required boundaries: coverage must be at least 90%, while MIS ratio must be below 1.00.

DATA AND DESIGN

Training data, validation and testing

Stage	Dates	How it is used
Prepared source	Jan 2010-Jun 2026	396 rows: 198 monthly observations for each of two HS series.
Expanding training	Starts Jan 2010	Each fold/origin uses only rows available before that forecast.
Blocked validation 1	Jan-Dec 2017	Model-family comparison and quantile calibration evidence.
Blocked validation 2	Jan-Dec 2018	Same frozen candidate comparison.
Blocked validation 3	Jan-Dec 2019	Same frozen candidate comparison.
Blocked validation 4	Feb 2020-Jan 2021	Includes pandemic-era behaviour before the main test.
Blocked validation 5	Feb 2021-Jan 2022	Final pre-test selection window.
Required test	Feb 2022-Dec 2023	23 monthly origins; disruption acceptance test.
Additional test	Jan 2024-Dec 2025	24 monthly origins; post-disruption robustness evidence.
Latest reference fit	Through Jun 2026	Produces current reference H3/H6/H12 forecasts.

Data used

Feature group	Examples and purpose
UK import price	UN Comtrade trade value and net weight; USD unit value converted to GBP/kg using Bank of England FX.
Price dynamics	GBP price changes over 1, 2, 3, 6 and 12 months; 6- and 12-month volatility.
Import quantity	Weight changes over 1, 3 and 12 months, volume volatility, low-volume and relative-volume indicators.
Supplier structure	USA, Japan, France, Russia, China and other shares; HHI, top supplier share, partner count and recent concentration change.
Currency	GBP/USD changes over 1, 3, 6 and 12 months and six-month FX volatility.
Industrial-price proxy	ERUC UK import PPI changes and volatility for other non-ferrous metals.
Commodity context	World Bank aluminium and nickel changes, with source-age and stale-data flags.
Titanium supply proxy	Annual USGS titanium sponge price and year-on-year change, with source-age and stale flags.
Seasonality and quality	Month sine/cosine, price-shock, estimated-weight, low-volume, EU collection-change and target-reliability flags.

Models considered

Weighted linear quantile regression, gradient-boosted quantile regression, Random Forest distribution, Elastic Net residual model, robust Huber residual model, and an adaptive no-change fallback. Model families are selected using only the five blocked pre-test validation windows.

READINESS

What remains before production

Current decision: not production approved. Use the forecasts in shadow mode with analyst review.

Priority	Pending improvement	Why it matters
1	Refresh World Bank and titanium supply inputs	Latest reference rows carry stale external-data flags; stale inputs weaken current forecasts.
2	Run a genuinely prospective holdout	2024-2025 was examined during development and cannot be treated as untouched final evidence.
3	Improve 810890 modelling	Processed titanium is heterogeneous; partner/product-mix shifts make a single unit-value series unstable.
4	Repair 810820 H3 stability	It passes the disruption test but MIS is 21.9% worse than baseline in 2024-2025.
5	Accumulate and validate H12 outcomes	H12 is report-only and later post-period targets are incomplete.
6	Add cross-horizon interval coherence	Current 810820 P95 and interval width shrink from H3 to H12 without explicit joint calibration.
7	Operational monitoring and gates	Automate source freshness, schema checks, drift alerts, missing-output checks and baseline comparison.
8	Consider HMRC as primary source	The specification prefers HMRC; UN Comtrade is an allowed fallback but may contain revisions and estimated weights.

Recommended deployment interpretation

- 810820 H6: strongest shadow candidate; passed both periods.
- 810820 H3: research-only until recent-period interval performance improves.
- 810890 H3/H6: analyst-reviewed fallback forecasts; disruption robustness is not proven.
- H12: planning reference only, not an accepted production forecast.

The two evaluations show that good coverage alone is insufficient. A production forecast must cover actual outcomes with useful intervals, beat a simple no-change benchmark, recover after shocks, and remain stable in later periods.

Evidence sources

Local Model A Production V3 outputs: stress_2022_2023/evaluation_metrics.csv; post_2024_2025/evaluation_metrics.csv; titanium_model_a_production_v3_training.csv; SCI_Model_Specifications_October_2026.md.